Ritesh Somashekar

Data Engineer | AWS, Snowflake & Python

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PROFESSIONAL SUMMARY

- **Data Engineer** with **8+ years of experience** building scalable **ETL/ELT pipelines** and cloud data platforms on **AWS** and **Snowflake** across technology, healthcare, research, and industrial environments.
- Hands-on with **AWS Glue, S3, Lambda, Athena, EMR,** and **EC2**, plus **Apache Airflow, Spark/PySpark,** and **Kafka**, delivering **99.9% pipeline reliability** with end-to-end lineage tracking, monitoring, and automated data quality checks.
- Experienced in **data modeling** and **warehousing** (**Snowflake, Databricks, Delta Lake**), **HIPAA-compliant** data handling, and partnering with analytics, ML, and business teams to deliver trusted, analysis-ready data.

TECHNICAL SKILLS

AWS Data Services: AWS Glue, **S3**, **Lambda**, **Athena**, **EMR**, **EC2**, IAM, SageMaker

Data Warehousing & Lakes: Snowflake, Databricks, Delta Lake, Azure Data Lake, Star Schema, Dimensional Modeling, Data Warehousing

Pipelines & Orchestration: ETL/ELT, **Apache Airflow**, Apache Spark, **PySpark**, Kafka, Flink, dbt, Azure Data Factory, Hadoop, Hive, Sqoop, Informatica PowerCenter

Programming & Databases: Python, **SQL**, Scala, Java, Bash, PostgreSQL, Oracle, SQL Server, MySQL, MongoDB

Data Quality & Operations: Data Validation, Data Lineage, Pipeline Monitoring & Alerting, Git, GitHub, CI/CD, Docker, Kubernetes, Linux

Analytics & ML: Power BI, Tableau, Pandas, Scikit-learn, MLflow, FastAPI, REST APIs

EDUCATION

Master's in Data Analytics and Engineering | George Mason University, Fairfax, VA, USA

Bachelor's in Information Science and Engineering | Visvesvaraya Technological University, Bengaluru, Karnataka, India

PROFESSIONAL EXPERIENCE

Apple | ***AI & ML Engineer*** **Cupertino, CA | May 2025 - Present**

- Designed large-scale data and feature engineering pipelines using **Python, Apache Spark,** and **AWS EMR**, processing behavioral and telemetry data for model development, performance tracking, and analytics.
- Engineered **real-time streaming pipelines** with **Apache Kafka, Flink,** and **Spark** to ingest and preprocess interaction data for downstream models, improving pipeline throughput by **26%**.
- Developed anomaly detection workflows using LightGBM, statistical evaluation, and threshold-based alerts, reducing false-positive alerts by **25%**.
- Automated data and model release workflows with **MLflow, FastAPI, Docker,** and **GitHub Actions**, reducing release cycles from two weeks to five days and improving reproducibility.

George Mason University | ***Graduate Research Assistant (Applied ML)*** **Fairfax, VA | August 2024 - Present**

- Built end-to-end data pipelines using **Python, SQL, Pandas, AWS S3,** and **DVC** to process **30,000 academic documents**, covering cleaning, deduplication, and metadata tagging; improved entity-recognition F1-score by **18%** over spaCy.
- Standardized **15GB** of structured and unstructured data across six datasets using Pandas, reducing preparation time by **30%** through repeatable transformation and validation.
- Led the data pipeline for the **SHIELD** and **MATE-911** AI platforms (published at **ACM WSDM '26** and **IEEE PST 2026**), turning unstructured 9-1-1 call records into **160** validated, embedding-ready scenarios.
- Built reproducible **Python** and MLflow pipelines with Optuna, reducing iteration cycles from three days to **under 18 hours**, and deployed data APIs with **FastAPI and Docker** at **sub-200ms** latency.
- Collaborated with research teams using **Git** and **DVC** to document data and analysis methods, track experiments, and maintain reproducible workflows.

Accenture | ***Machine Learning Engineer*** **India | September 2020 - July 2023**

- Designed and managed **20+ enterprise ETL pipelines** on **AWS**, boosting data throughput by **70%** and ensuring high availability for downstream AI and BI applications.
- Automated workflows using **Apache Airflow, Python,** and **AWS (S3, Lambda, Athena)**, achieving **99.9% pipeline reliability** with end-to-end lineage tracking.
- Integrated structured and unstructured healthcare data with **PySpark, AWS Glue, S3,** and **Delta Lake**, building ETL pipelines that improved data reliability by **30%** under **HIPAA** requirements.
- Engineered ingestion pipelines for **400K+ XML/JSON healthcare records** into **Snowflake** and **Databricks**, accelerating model training by **60%** and reducing preprocessing time by **70%**.
- Built a daily ingestion workflow from **NPPES** repositories, improving data accuracy by **74%** and cutting processing overhead by **60%**.
- Engineered **PySpark** and **Databricks** pipelines processing **500GB+** of structured and unstructured data, reducing processing time by **45%** for downstream ML applications.
- Delivered self-service BI pipelines integrating **Tableau, Power BI,** and SQL-based models, reducing reporting latency by **40%.**
- Partnered with data engineering, product, and client stakeholders to define KPIs, evaluate solution performance, and prioritize delivery, reducing turnaround time by **15%.**

Honeywell | ***Data Engineer*** **India | January 2018 - September 2020**

- Engineered **ETL/ELT pipelines** for industrial IoT analytics with **Python, PySpark,** and **Apache Spark**, ingesting telemetry from **8+ source systems** via REST APIs and processing **2M+ records daily** into **Azure Data Lake**.
- Automated batch workflows with **Azure Data Factory** and **Apache Airflow**, adding source-to-target validation and monitoring across **15+ daily jobs** and cutting issue detection time to **under 15 minutes**.
- Implemented **data quality frameworks** (null checks, duplicate detection, schema validation, referential integrity), reducing downstream reporting errors by **40%** across **3 plant sites**.
- Designed a **star-schema** data warehouse for asset uptime and operational KPIs, cutting report turnaround time by **60%** across **5 operational teams**.
- Migrated large-scale workloads from **Hive** to **Apache Spark/PySpark**, and moved **Hadoop/Sqoop** ingestion to **Azure Data Services**, improving performance and scalability.
- Archived raw telemetry in **AWS S3**, designed **MongoDB** models for high-volume sensor data, and managed deployments with **Git, CI/CD,** and Linux shell scripting.